

first message to a new large group, or similar large scale sets of synchronizations being required at a single point in time.

### 4.18.2. Group Peer State

The *Group Peer State Table* stores information about every peer with which the node had a group message exchange. For every peer node id the following information is available in the table:

- Peer's Encrypted Group Data Message Counter Status:
  - *Synchronized Data Message Counter* - the largest encrypted data message counter received from the peer, if available.
  - Flag to indicate whether this counter value is valid and synchronized.
  - The *message reception state* bitmap tracking the recent window of data message counters received from the peer.
- Peer's Encrypted Group Control Message Counter Status:
  - *Synchronized Control Message Counter* - the largest encrypted control message counter received from the peer, if available.
  - Flag to indicate whether this counter value is valid and synchronized.
  - The *message reception state* bitmap tracking the recent window of control message counters received from the peer.

There are three scenarios where the receiver of an encrypted message does not know the sender's last message counter:

- The encrypted message is the first received from a peer.
- The device rebooted without persisting the *Group Peer State Table* content.

**NOTE** It is not required to persist the Peer State Table.

- The entry for the Peer in the *Group Peer State Table* was expunged due to the table being full.

There SHALL be at least 10 entries per supported fabric for Peer Encrypted Group data Message Status in the Group Peer State table. This number SHOULD NOT be less than 5 entries per fabric per instance of Groups cluster on an endpoint. The number of entries is a compromise between the number of peers that can send a group message to a receiver while the table is maintained, and the associated memory usage.

There SHALL be at least 2 entries per supported fabric for Peer Encrypted Group Control Message Status in the Group Peer State table.

The next sections describe the functional protocol used to request message counter synchronization with a peer and form responses to such message counter synchronization requests.

### 4.18.3. MCSP Messages